

Arizona State University

IEE572: Design of Experiments

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Project Report 3

Experiment: Non-stop Driving

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1. Purpose

The purpose of this document is to capture the first three planning steps for designing the Non-stop Driving experiment. This study aims to identify the factors that allow a daily commute to be completed with the minimum number of stops at traffic lights, thereby improving fuel efficiency and reducing travel time.

2. Problem Statement – Step 1

2.1 Background

The driver operates a standard passenger vehicle on a set urban or suburban commuting route. Traditional commuting often involves frequent idling and braking due to poorly synchronized traffic signals or inconsistent driving speeds, which leads to increased wear on the vehicle and higher stress for the driver. Modern vehicles and navigation tools provide data that could potentially be used to optimize "flow," but the specific combination of driver-controlled variables that lead to a "non-stop" commute remains unquantified.

2.2 Goal

Identify the ideal driving parameters (such as departure time, speed, and lane choice) to maximize the probability of a commute being completed without coming to a full stop at a traffic light.

3. Variable Selection (Factors, Levels, and Ranges) – Step 2

To maximize the probability of a smooth commute, and avoiding stopping at traffic lights, several independent variables are taken into consideration. Such variables are crucial to be defined at this stage of the experiment, as they represent the primary conditions that can affect the driving journey. The defined independent variables will vary the factors mentioned below in Table 1. These variables are chosen since they are highly affecting the driver's journey between consecutive traffic lights. The table below demonstrates the factors, levels and ranges of variation.

Table 1: Varied Factors

#	Factor	Units	Level 1 (Low)	Level 2 (High)	Ranges
1	Average driving speed	mph	35 mph	50 mph	35-50 mph
2	Traffic lights count	Count	5 lights	12 lights	5-12 lights
3	Departure Time	Time (AM)	7:00 AM	10:00 AM	7:00-10:00 AM
4	Day of the Week	Categorical	Monday	Wednesday	Weekdays
5	Navigation Assistance Mode	Category	Off	On (live traffic enabled)	With/Without Navigation

Explanation of the chosen Factors:

1. Average Driving Speed: the driver's speed can determine how fast the vehicle can approach each traffic light's intersection. Variation in the overall speed will determine the probability of whether the driver will reach the light when it is green or red.
2. Traffic Lights Count: routes and streets with more traffic lights will affect and increase the probability of encountering a full stop at red signals.
3. Departure Time: during peak and off-peak hours, there are notable differences in traffic volume and signal coordination efficiency.
4. Day of the Week (blocking factor): the traffic density and patterns vary depending on the day of the week – due to different commuting schedules related to work and school, and also the generic daily traffic.
5. Navigation Assistance Mode: drivers can be rerouted past traffic jams or areas with a lot of signals using real-time guidance apps.

4. Responses – Step 3

This experiment's goal is to examine and measure how certain controllable conditions can affect how easy a daily journey is, as indicated earlier in the problem statement above.

4.1 Response Variable

Table 2: Experimental Response Variable

Response	Description	Units	Aim
Number of Full Stops at Traffic Lights	Measure the number of times the vehicle stops fully at a traffic lights	Count	Minimize

4.2 Fixed Factors

To ensure experimental consistency, the below table shows the factors that will remain constant throughout all trials.

Table 3: Fixed Factors & Conditions

Factors	Fixed Conditions
Vehicle	Same vehicle for all experimental runs
Driver	Same driver for all runs
Origin and Destination	Identical starting and ending locations
Weather Condition	Clear and dry conditions only
Data Collection Method	Same device and recording method for each trial
Traffic Law Compliance	Full adherence to traffic regulations

5. Experimental Design – Step 4

The following part of the report shows the experiment design chosen by the team, to complete this experiment. Different sections are discussed in this section, starting from the chose factors of interest until the JMP table is generated.

5.1 Experimental Design Selection

The design selected for this experiment is the full factorial Design of Experiments (DOE), using the four primary factors, where each factor is at two levels. As for the 5th factor (day of the week), it is chosen to be a blocking factor. Therefore, this results in a 2^4 full factorial design. And so, this will give us $2^4 = 16$ **experimental runs per block**.

The chosen four factors of interest are as follows:

1. Average driving speed
2. Traffic lights count
3. Departure Time
4. Navigation Assistance Mode

5.2 Blocking Factor

In addition to the factors of interest, “Day of the Week” is set as a blocking factor. This is due to the significant effect this factor can have on the main aim of the experiment’s response, where its variability is not the goal of this study.

This particular blocking factor has two levels, set as Monday and Wednesday. Therefore, by combining both the experimental runs per block with these two blocks, this results in more accurate experimental results, as follows:

16 experimental runs/block X 2 block = 32 experimental runs

5.3 Experimental Design Matrix

The experimental design matrix was created and designed using JMP. The purpose of generating a JMP table is to combine and group all the four factors of interest at two levels each, resulting in 16 runs per block. The following table, Table 4, illustrates the JMP table with all 16 unique experimental conditions:

Table SEQ Table * ARABIC 4: JMP Table – Experiment Run Sheet

	Pattern	Avg Driving Speed	Traffic Lights Count	Departure Time	Navigation Assistance Mode	Number of Stops	Day
1	---+1	35	5	10	Off		Monday
2	++-1	50	12	7	Off		Monday
3	+--1	50	5	7	Off		Monday
4	---1	35	5	7	Off		Monday
5	+++1	50	12	10	Off		Monday
6	-++2	35	12	7	On		Monday
7	+-+1	50	5	10	Off		Monday
8	-+-1	35	12	7	Off		Monday
9	+++2	50	12	10	On		Monday
10	---2	35	5	7	On		Monday
11	++-2	50	12	7	On		Monday
12	+-+2	50	5	10	On		Monday
13	+--2	50	5	7	On		Monday
14	-++2	35	12	10	On		Monday
15	-++1	35	12	10	Off		Monday
16	---2	35	5	10	On		Monday
17	---+1	35	5	10	Off		Wednesday
18	++-1	50	12	7	Off		Wednesday
19	+--1	50	5	7	Off		Wednesday
20	---1	35	5	7	Off		Wednesday
21	+++1	50	12	10	Off		Wednesday

2 2	---2	35	12	7	On		Wednesd ay
2 3	++1	50	5	10	Off		Wednesd ay
2 4	---1	35	12	7	Off		Wednesd ay
2 5	+++2	50	12	10	On		Wednesd ay
2 6	---2	35	5	7	On		Wednesd ay
2 7	++2	50	12	7	On		Wednesd ay
2 8	++2	50	5	10	On		Wednesd ay
2 9	---2	50	5	7	On		Wednesd ay
3 0	---2	35	12	10	On		Wednesd ay
3 1	++1	35	12	10	Off		Wednesd ay
3 2	---2	35	5	10	On		Wednesd ay

6. Performing the Experiment – Step 5

6.1. Experimental Procedure

Each run represents a unique set of the four selected factors: average driving speed, traffic lights count, departure time, and navigation assistance usage. Moreover, each run is conducted across two blocks, representing two different days of the week, to help interpret the variability in traffic conditions during the week.

6.2. Data Collection

The aim of this experiment is to measure the number of full stops at traffic lights, which was set to be the response variable of this experiment. A full stop at traffic light is defined as any instance the vehicle comes to a complete stop due to the traffic a at the given factors (due to vehicle speed, number of traffic lights,

time of commute, and other factors that would impact the smoothness of the trip. Each experimental run will be captured in table 4, where it corresponds to the response variable according to the experimental design.

6.3. Data Simulation – Response Generation

The responses of this experiment were simulated due to the practical constraints that might arise from real traffic conditions. And so, a well-defined dataset structure was generated, by comparing it to actual day-to-day driving, to simulate and generate the response variable. The values were simulated based on actual traffic behavior and realistic judgement of the group members. The below points illustrate the methodology used to help in judging and deciding on each dataset:

- Low driving speeds may result in a greater number of stops.
- More traffic lights may result in a greater number of stops.
- Early departure time (during peak hours) may result in a greater number of stops.
- Using GPS navigation (with live traffic updates) will reduce the number of stops.
- Variations between Monday and Wednesday were measured to analyze the differences in traffic conditions.

6.4. Data Recording

This section of the report shows the JMP table designed by the team, to define each unique use case by combining different factors and ranges as given. Therefore, the responses were captured and added, as shown in Table 5 below:

	Pattern	Avg Driving Speed	Traffic Lights Count	Departure Time	Navigation Assistance Mode	Number of Stops	Day
1	--+1	35	5	10	Off	4	Monday
2	++-1	50	12	7	Off	7	Monday
3	+--1	50	5	7	Off	4	Monday

4	---1	35	5	7	Off	5	Monday
5	+++1	50	12	10	Off	6	Monday
6	+-+2	35	12	7	On	7	Monday
7	+--+1	50	5	10	Off	3	Monday
8	-+-1	35	12	7	Off	8	Monday
9	+++2	50	12	10	On	5	Monday
10	---2	35	5	7	On	4	Monday
11	++-2	50	12	7	On	6	Monday
12	+--+2	50	5	10	On	1	Monday
13	+--2	50	5	7	On	2	Monday
14	-++2	35	12	10	On	6	Monday
15	-++1	35	12	10	Off	7	Monday
16	--+2	35	5	10	On	3	Monday
17	--+1	35	5	10	Off	3	Wednesday
18	++-1	50	12	7	Off	6	Wednesday
19	+--1	50	5	7	Off	3	Wednesday
20	---1	35	5	7	Off	4	Wednesday
21	+++1	50	12	10	Off	5	Wednesday
22	+-+2	35	12	7	On	6	Wednesday
23	+--+1	50	5	10	Off	2	Wednesday
24	-+-1	35	12	7	Off	7	Wednesday
25	+++2	50	12	10	On	4	Wednesday
26	---2	35	5	7	On	3	Wednesday
27	++-2	50	12	7	On	5	Wednesday
28	+--+2	50	5	10	On	0	Wednesday
29	+--2	50	5	7	On	1	Wednesday
30	-++2	35	12	10	On	5	Wednesday
31	-++1	35	12	10	Off	6	Wednesday
32	--+2	35	5	10	On	2	Wednesday

7. Statistical Analysis of Data – Step 6

7.1. Analysis Approach

The findings from Table 5 were analyzed using JMP, where a full factorial model was used to evaluate and analyze the effects of the chosen factors on the number of stops. The below parts of this section include some of the important JMP results – such as, statistical analysis (ANOVA), main effects, and interaction effects. All plots shown below were generated using JMP.

7.2. Anova Analysis

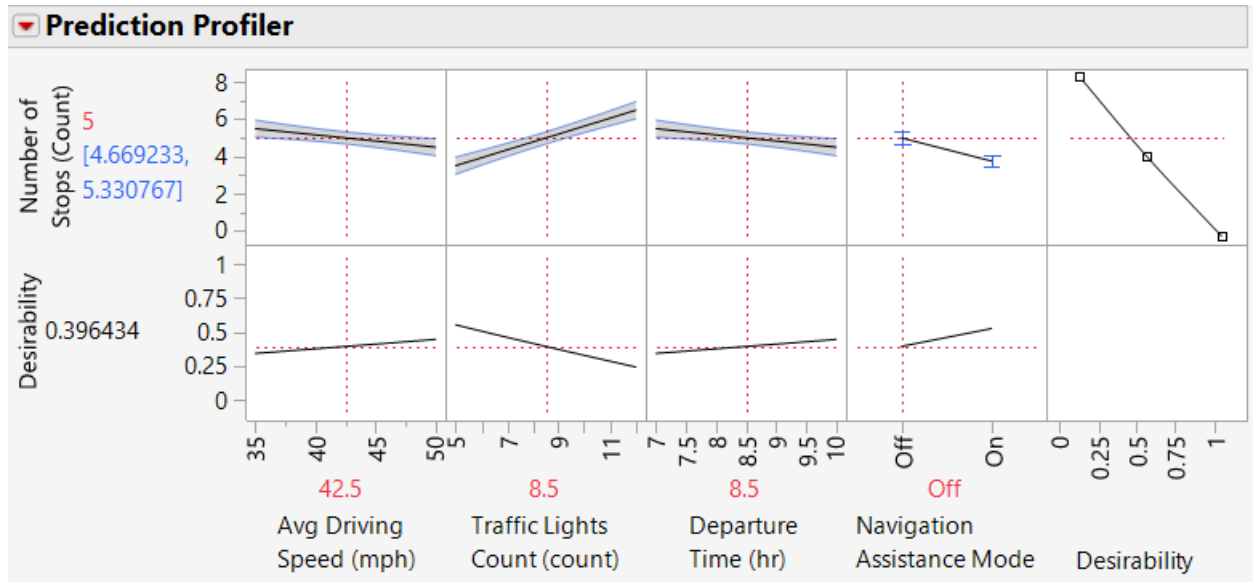
Once a full factorial analysis is run on the dataset, the effect analysis

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Avg Driving Speed(35,50)	1	1	12.500000	30.8824	<.0001*
Avg Driving Speed*Departure Time	1	1	0.000000	0.0000	1.0000
Avg Driving Speed*Navigation Assistance Mode	1	1	0.500000	1.2353	0.2789
Avg Driving Speed*Traffic Lights Count	1	1	0.500000	1.2353	0.2789
Departure Time(7,10)	1	1	8.000000	19.7647	0.0002*
Departure Time*Navigation Assistance Mode	1	1	0.000000	0.0000	1.0000
Navigation Assistance Mode	1	1	12.500000	30.8824	<.0001*
Traffic Lights Count(5,12)	1	1	84.500000	208.7647	<.0001*
Traffic Lights Count*Departure Time	1	1	0.000000	0.0000	1.0000
Traffic Lights Count*Navigation Assistance Mode	1	1	0.500000	1.2353	0.2789

(ANOVA) was generated. Figure 1 presents the results of the ANOVA, with the different combination of the factors as shown. The p-values (Prob > F) of the four main factors – avg driving speed, traffic lights count, departure time, and navigation assistance mode – indicates that they are all statistically significant due to their p-values being less than 0.05.

Of the four main factors, the traffic lights count demonstrated the strongest effect, this is reflected by having the highest Sum of Squares and F Ratio, as shown in Figure 1. Furthermore, different interaction effects were captured from the ANOVA results, such as the combination that includes traffic lights count and GPS, this indicated their impact on the overall experimental factors when compared to others.

7.3. Main Effect Analysis

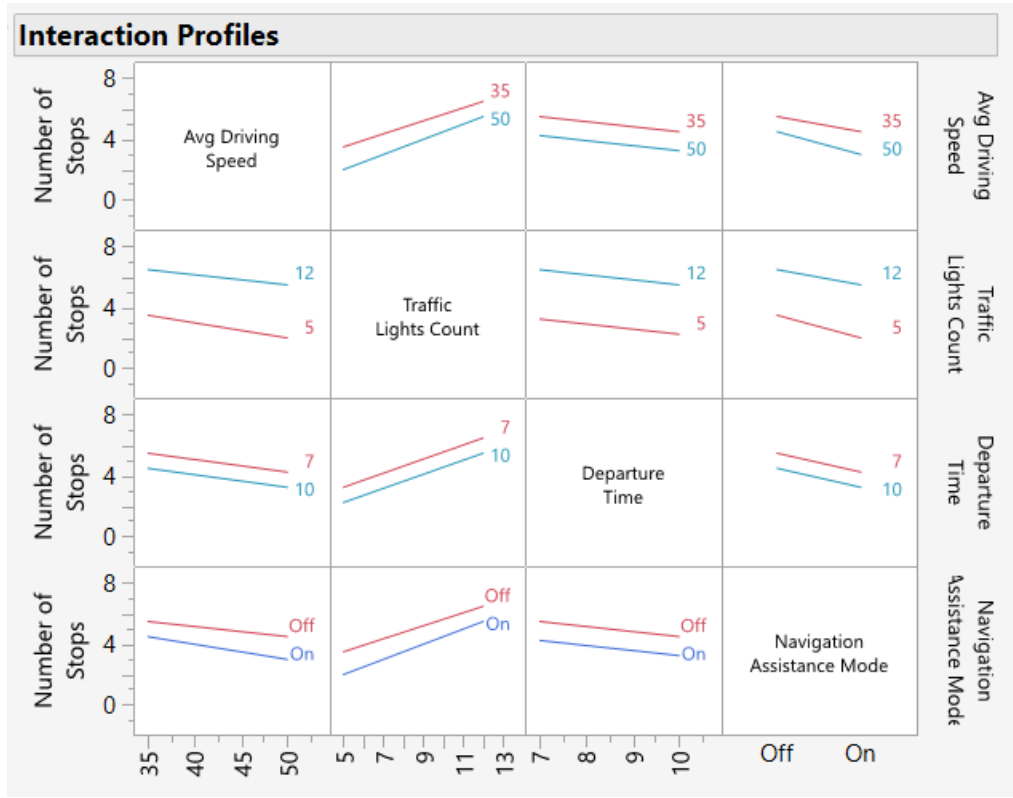


The above plots shown in Figure 2 show the main effects of the chosen factors on the number of stops. From the performed analysis, a prediction profiler will be generated. Some of the takeaways from these plots include the following:

- Driving at the speed limit (the higher range limit – 50 mph) results in less stops.
- More traffic lights in the route lead to higher possibility of stopping at each traffic light.
- Early departure results in a greater number of stops, due to the daily traffic congestion.
- Using a live navigation system reduces the number of stops, due to improved route efficiency and traffic monitoring.

As a result of these plots, it can be concluded that the steep slopes captured for traffic lights count and avg driving speed proves their effect on impacting the response variable, where the driver stops fully at the majority of the traffic lights.

7.4. Interaction Analysis



The above interaction profile plots, as shown in Figure 3, show the relationships between the main factors and how they can control the response variable, the number of stops.

From the generated plots, a strong interaction between average driving speed and traffic lights count is observed, which leads to the effect of how speed is dominated when the traffic lights count is low. Another significant interaction is the relationship between traffic lights count and the usage of the navigation assistance mode – this is due to the benefit of using a live traffic monitoring system.

Another conclusion that can be observed from the generated interaction plots, interactions that involve departure time tend to have a minimal effect on the overall experiment. This is indicated by the closely gapped parallel lines in the above plots – this suggests that the time's effect is almost independent of other factors.

7.5. Blocking Analysis

The blocking factor chosen for this experiment is the Day of the Week, where its varied between Monday and Wednesday, to account for and consider the variability in traffic conditions between the beginning and middle of the week.

From the ANOVA results, generated by JMP, the blocking factor showed a statistically significant effect with a p-value less than 0.05. This indicates how the variation in the day of the week has an effect on the response variable. Such conclusion can be also based on how the work nature and schedule of the employees varies during the week, and so this will impact on the overall experiment. For example, Monday commonly results in a higher number of stops per trip, when compared to Wednesday. This reflects the general and commonly increased traffic congestion at the beginning of the week.

As a result of generated analysis, it is proven that setting the Day of the Week as blocking factor resulted in a significant effect – confirming that it was appropriate and how it helped reduce the impact of external noise variability. And so, this allowed for a clearer assessment of the main experimental objective.

8. Conclusions and Recommendations – Step 7

8.1. Conclusions

The results of the Design of Experiments demonstrate that both route characteristics and driving behavior significantly influence the number of stops encountered during a commute. Among all factors, traffic lights count was identified as the most dominant factor, with a strong positive effect on the number of stops. This indicates that routes with a higher number of signalized intersections significantly increase the likelihood of stopping. In addition, average driving speed and departure time were found to be important factors. Higher driving speeds were associated with fewer stops, while earlier departure times (peak hours) resulted in increased stopping due to higher traffic congestion.

The use of GPS navigation also contributed to reducing the number of stops, suggesting that route optimization and real-time traffic information improve commute smoothness. Furthermore, the blocking factor (day of the week) was statistically significant, confirming that traffic conditions vary between different days. This validates the use of blocking in the experimental design and highlights the importance of accounting for external variability.

Overall, the experiment confirms that both controllable driving parameters and external conditions play a critical role in determining commute efficiency.

8.2. Recommendations

Based on the experimental data and conclusions obtained from this study, the following recommendations are given to help drivers reduce the amount of full stops throughout their daily travel :

First, drivers should prioritize routes that have fewer signalized junctions wherever possible. Avoiding high-signal routes can significantly lower stopping frequency and increase overall commute efficiency because the number of traffic lights was found to be the most significant factor in this investigation.

Second, it is recommended that you maintain a greater average driving speed while remaining inside the legal speed limit. The findings demonstrated that, in comparison to 35 mph, driving closer to 50 mph resulted in fewer stops. This is probably due to the fact that maintaining a higher speed increases the likelihood of arriving at crossings during green light periods.

Third, drivers should consider moving their departure time away from peak morning hours. The statistics consistently demonstrated that a 10:00 AM departure offered better results, while a 7:00 AM departure led to more stops because of increased traffic congestion. Where work schedules allow for flexibility, leaving later can make a significant difference.

Fourth, turning on live navigation aid is strongly advised. The findings revealed that GPS navigation with real-time traffic updates lowered stop counts across a

variety of run situations, notably on routes with increased traffic light density. This demonstrates the benefit of using a live traffic monitoring system during daily travels.

Finally, future studies should examine broadening the range of evaluated parameters, such as including weather conditions as a variable rather than fixing them, or testing extra days of the week to provide a more complete view of traffic variations throughout the week.